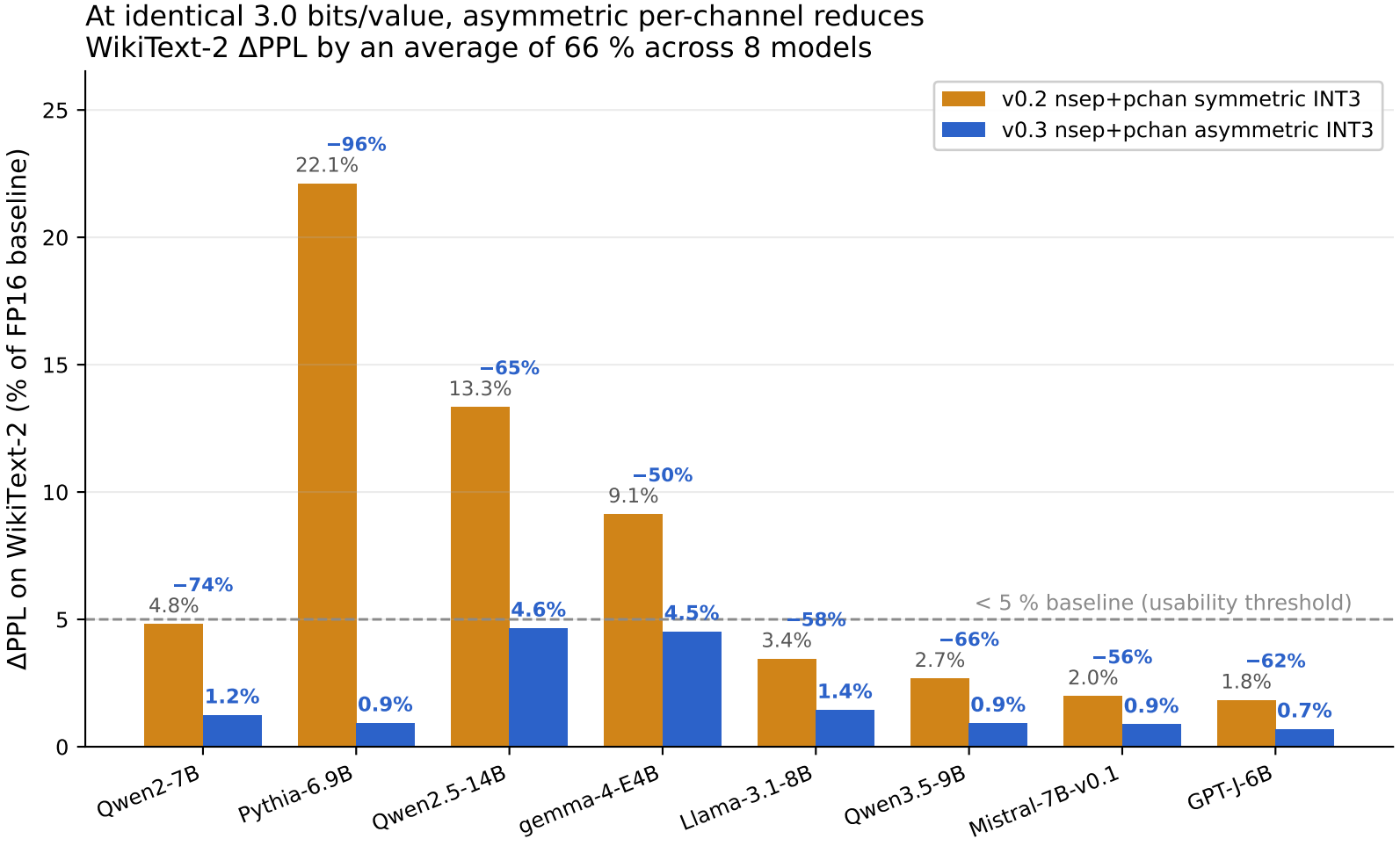
At identical 3.0 bits/value, asymmetric per-channel reduces
WikiText-2 ΔPPL by an average of 66 % across 8 models
v0.2 nsep+pchan symmetric INT3
v0.3 nsep+pchan asymmetric INT3
ΔPPL on WikiText-2 (% of FP16 baseline)
0
5
10
15
20
25
< 5 % baseline (usability threshold)
4.8%
−74%
1.2%
22.1%
−96%
0.9%
13.3%
−65%
4.6%
9.1%
−50%
4.5%
3.4%
−58%
1.4%
2.7%
−66%
0.9%
2.0%
−56%
0.9%
1.8%
−62%
0.7%
Qwen2-7B
Pythia-6.9B
Qwen2.5-14B
gemma-4-E4B
Llama-3.1-8B
Qwen3.5-9B
Mistral-7B-v0.1
GPT-J-6B